



NC STATE
UNIVERSITY

NRP 
Nuclear Reactor Program

NE235 Nuclear Reactor Operations Module #3

The “Three R’s”: Radioactivity, Radiation & Risk

Dept. of Nuclear Engineering
North Carolina State University

NE235 Nuclear Reactor Operations Training

Module #3

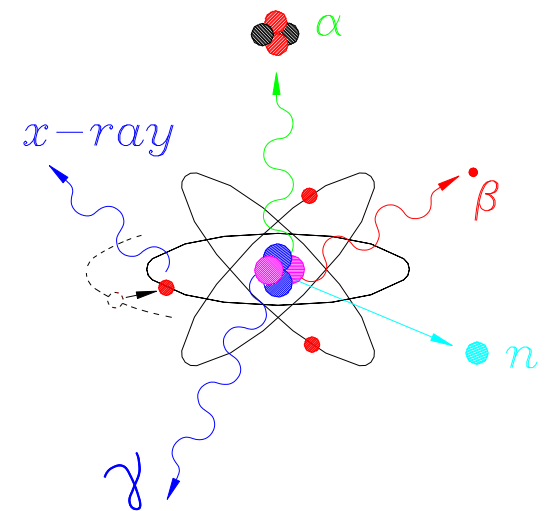
Module #3 Contents:

1. Radioactivity & reactions producing radioactivity inside reactor.
2. Nuclear radiation – Alpha, Beta, Gamma, Neutron
3. Radiation Dose & Biological Effects
4. Radiation Safety & Regulatory Limits (10CFR20)
5. Risk –vs- Radiation Dose

Laboratory Session this week:

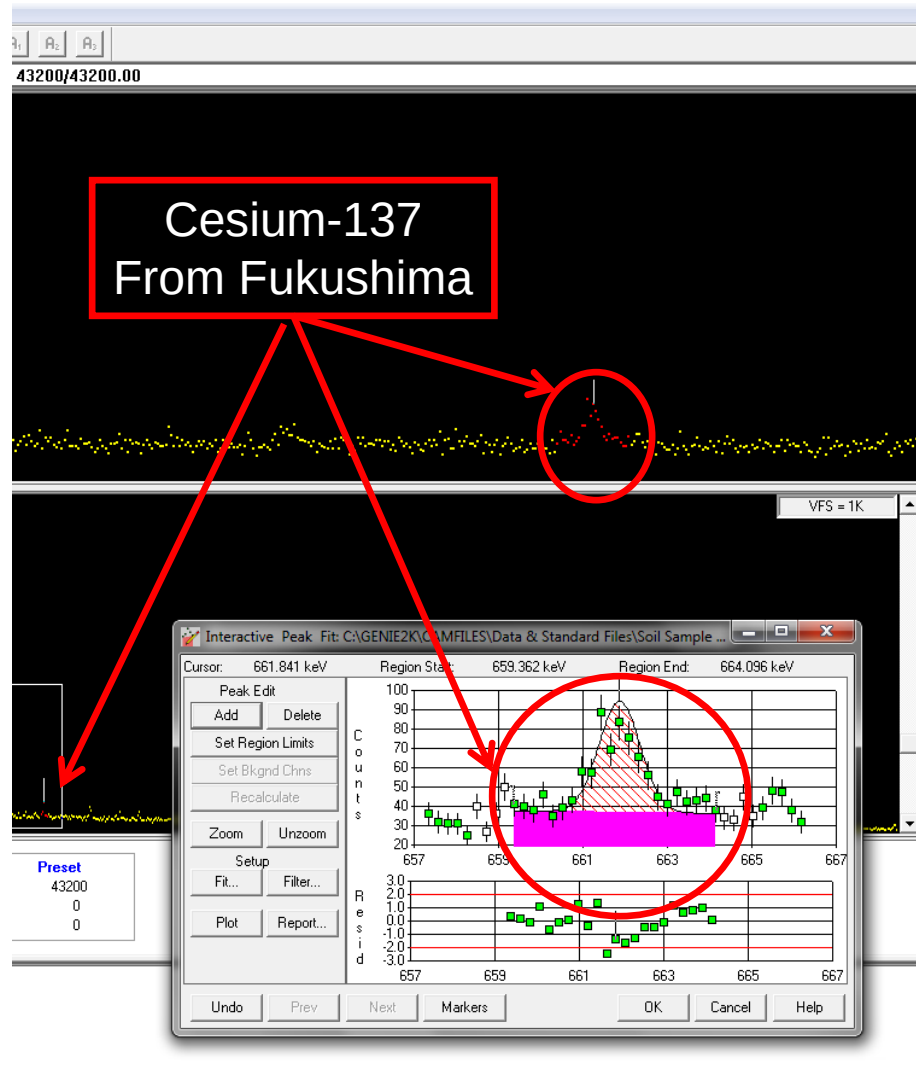
- Health Physics Lab on 9/8 (Burlington)

Homework #3 – Due 9/13



Fukushima Fallout

Sample taken Thursday April 7th, 2011 from storm drain in NCSU Brickyard

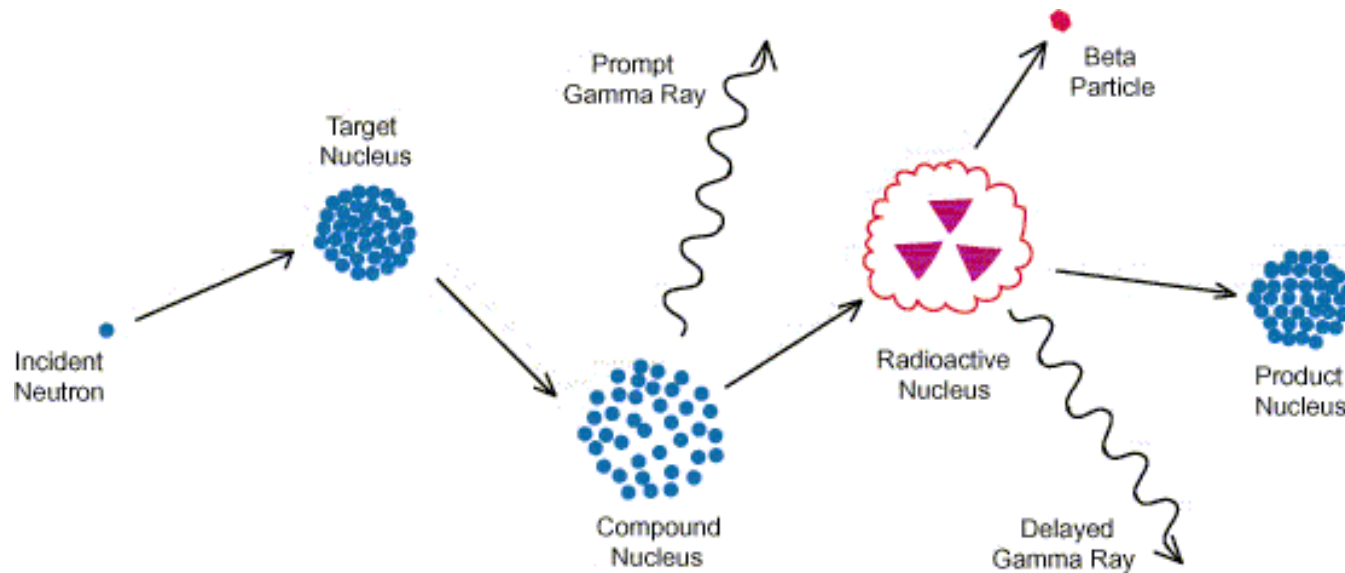


FACT:

You are currently being exposed to Cesium-137 fallout from Fukushima.

QUESTIONS:

- How much radioactivity is this?
- How much risk is associated with the dose resulting from exposure to this amount of radioactivity?
- Should you be worried??



Neutron Reactions in a Fission Reactor:

- Absorption reactions: **Capture, Fission, Breeding**
- Scattering reactions: **Moderation, Leakage, Reflection**

Radioactive materials in reactor lab:

- Reactor Fuel – fission γ 's & ^1_0n 's, fission products (^{135}I , ^{90}Sr , ^{140}La , etc...)
- Activation Products – ^{24}Na in coolant, ^{28}Al in core structures, ^{41}Ar in air
- Sealed Isotopic Sources - ^{60}Co , ^{137}Cs , ^{241}Am , $^{239}\text{Pu-Be}$

Radio - “Activity”

The **rate** at which radioactive material is decaying and releasing **ionizing radiation** is the **“Activity”**.

Units of Activity

- Becquerel (Bq):

$$1 \text{ Bq} = 1 \text{ DPS (decays per second)}$$

- Curie (Ci), mCi, μCi :

$$1 \text{ Ci} = 3.7 \times 10^{10} \text{ DPS} = \text{Activity in 1 gram of Radium}$$

Radioactive Decay Equation

$$A(t) = A_o e^{-\lambda t}$$

Where: A_o = *Initial Activity (DPS or Curies)*

$A(t)$ = *Activity after decay time “t”*

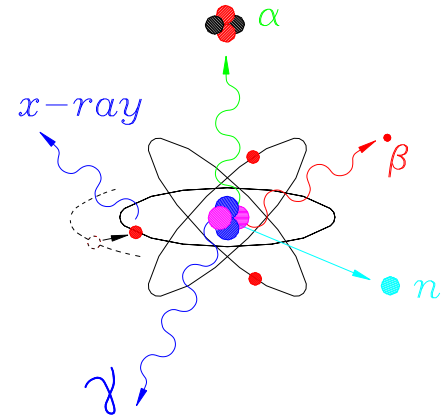
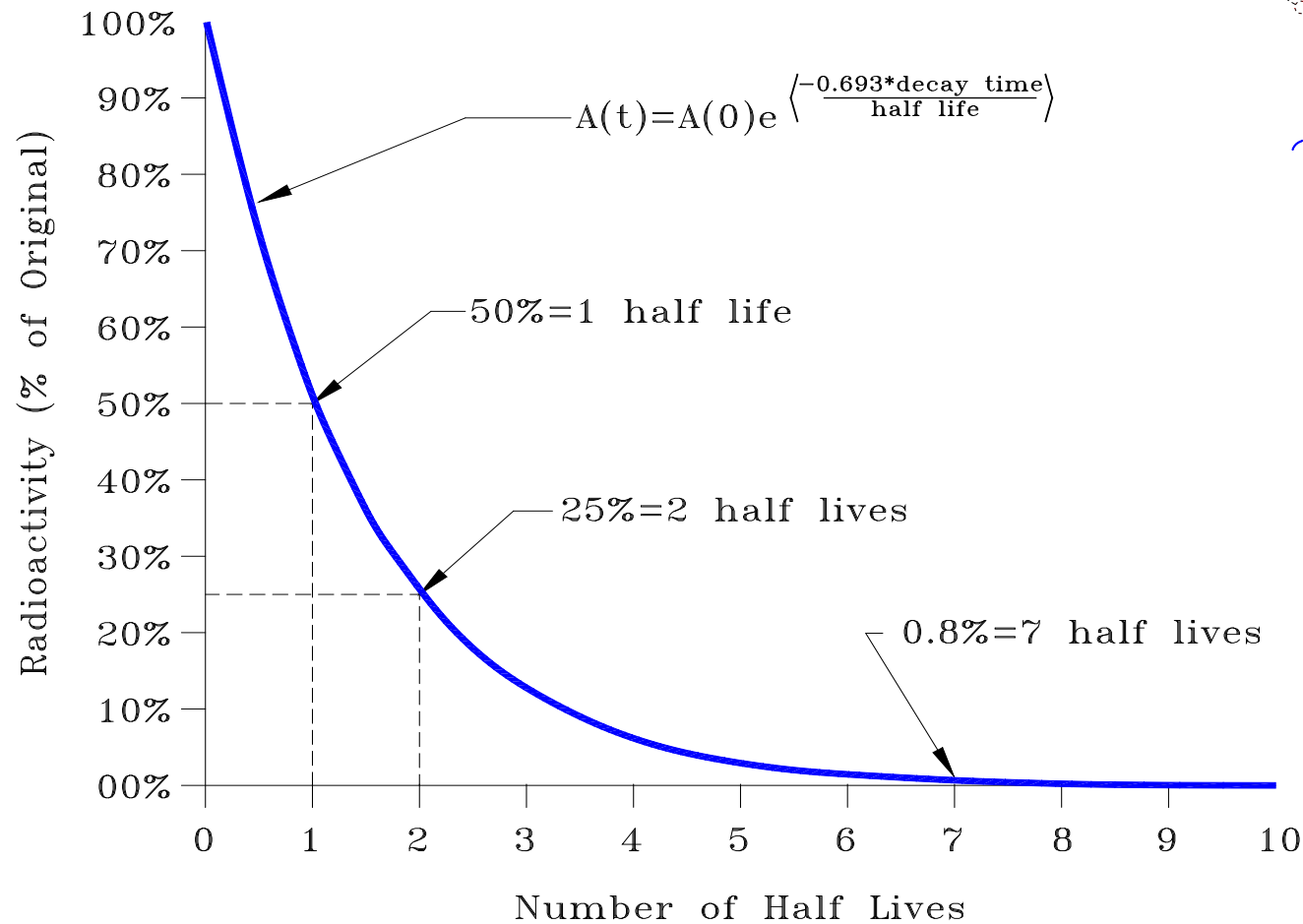
$\lambda = \ln(2) / T_{1/2}$

$T_{1/2}$ = *isotopic half life*

★ *The amount of radioactive material (activity A) decreases over time at a rate depending on the half life ($T_{1/2}$) of the decaying isotope.*

Radioactive Decay

- Function of isotopic half life



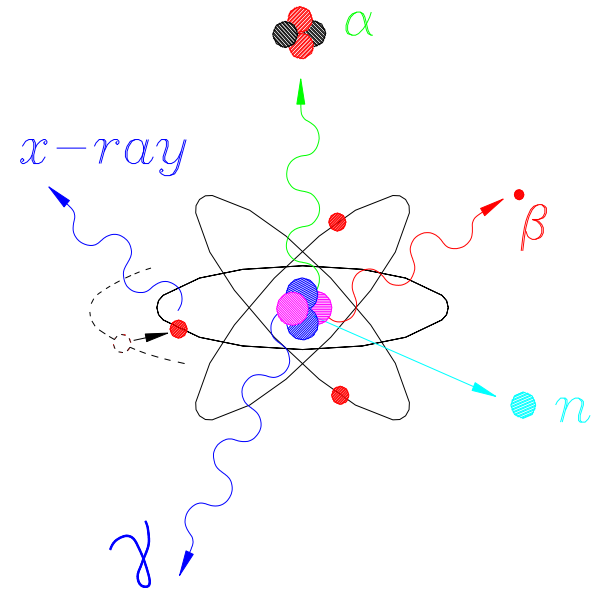
Ionizing –vs- Non-ionizing Radiation

❖ IONIZING RADIATION

- Particulate or Electromagnetic
- Charged: α^+ , β^-
- Uncharged: γ , X, n
- Causes ionization of atoms or molecules by breaking electron bonds
- breaks molecular bonds changing absorbing material chemistry

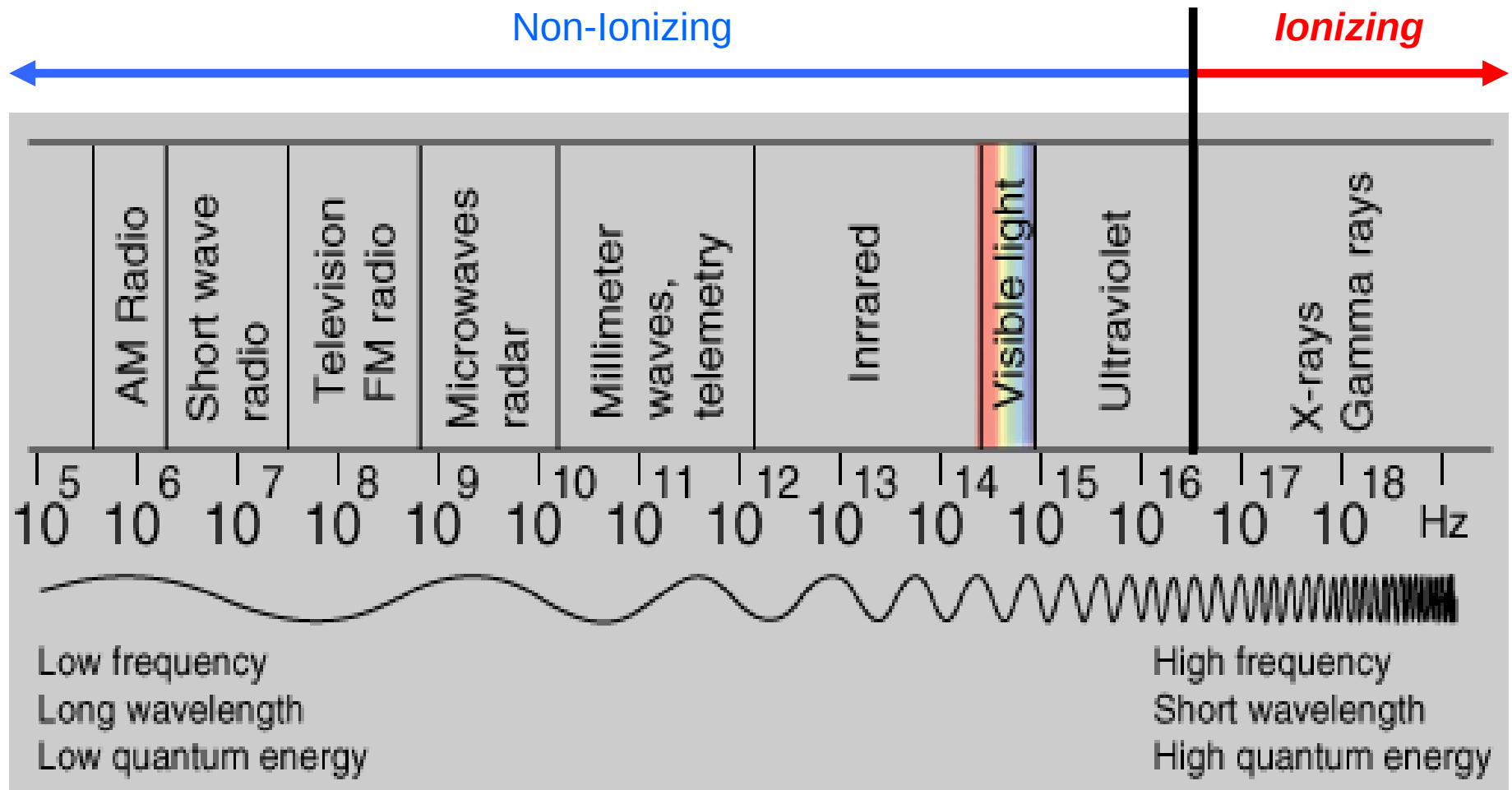
❖ NON-IONIZING RADIATION

- Electromagnetic (UV, IR, MW, RF)
- Can not ionize atoms or molecules – not enough energy to break electron bonds.

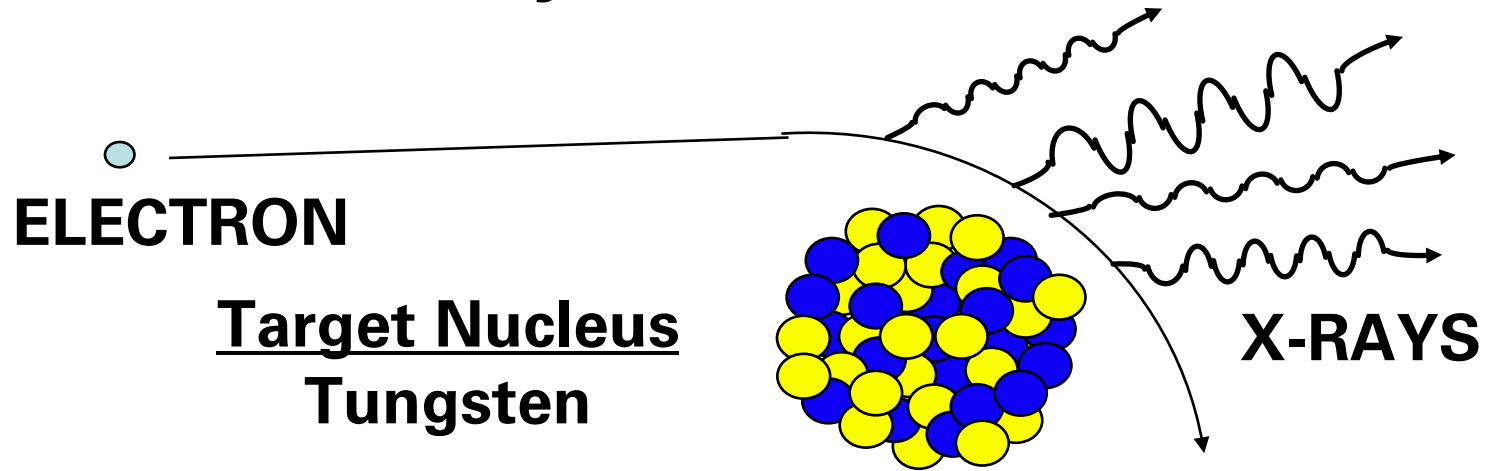


Electromagnetic Spectrum

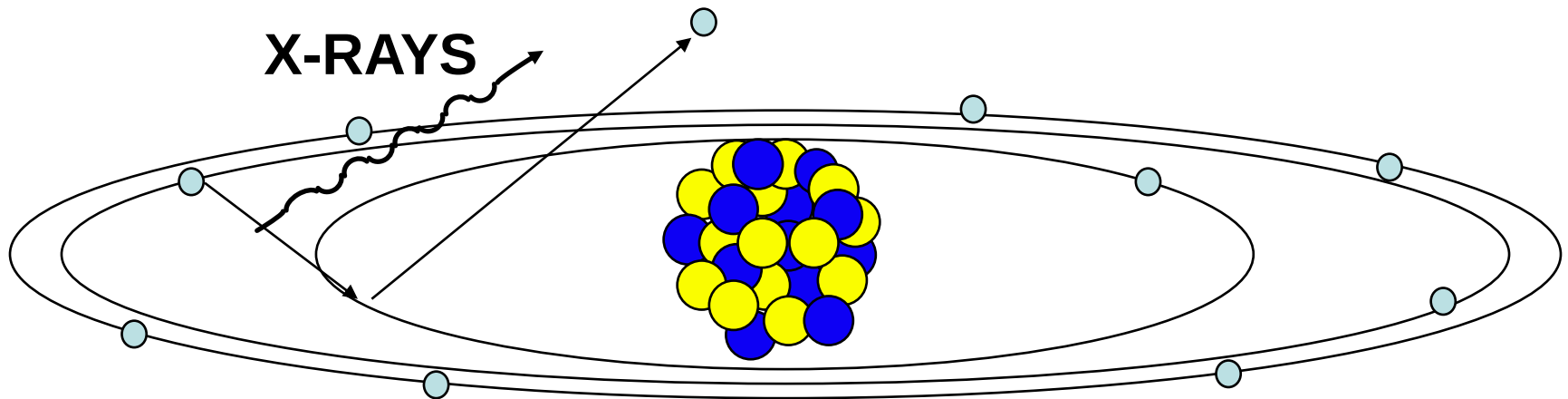
$$E = h\nu$$



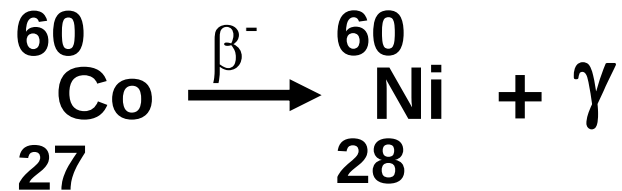
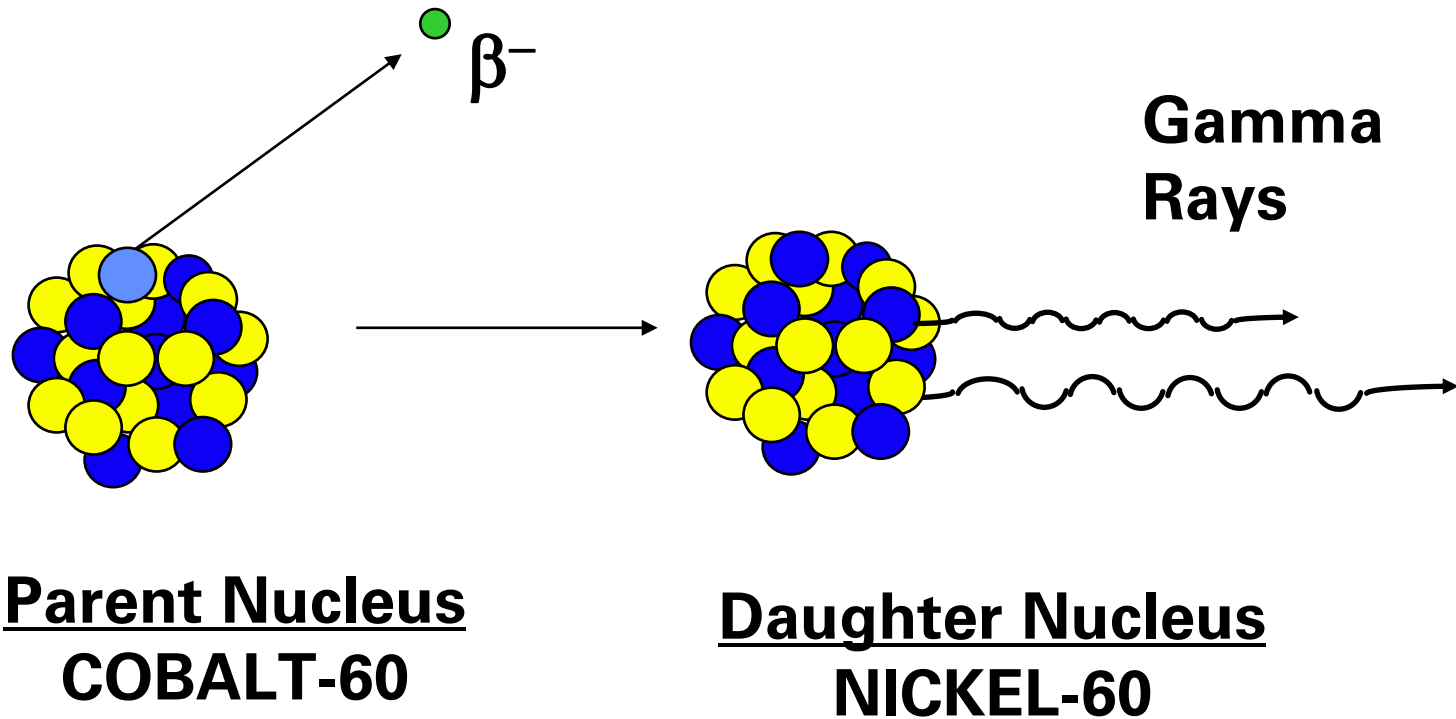
X-ray Production



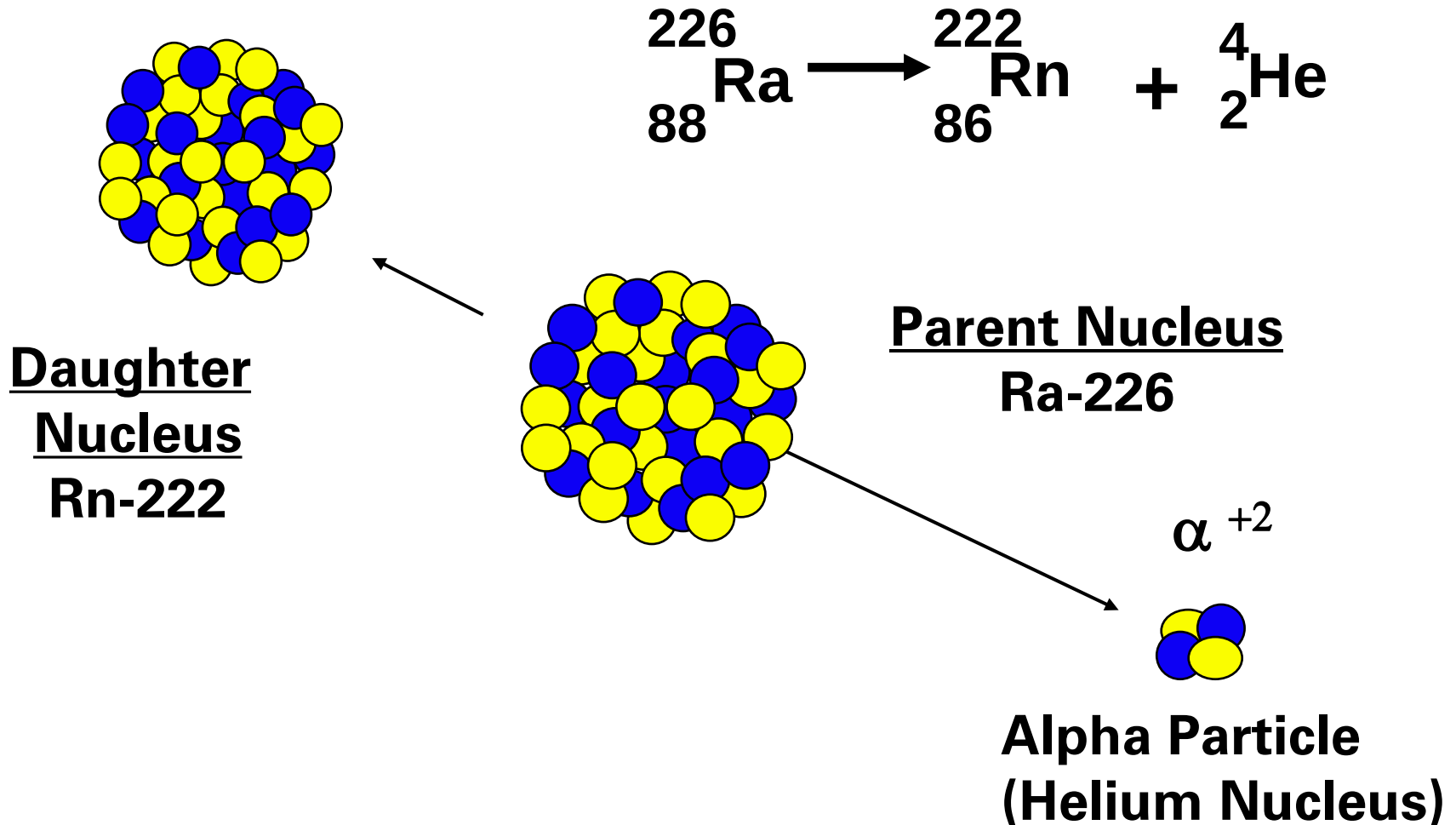
(BREMSSTRAHLUNG)



Gamma-ray (γ)



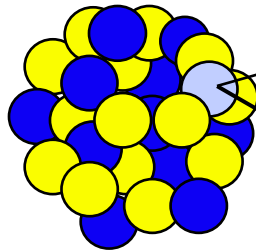
Alpha Particle (α)



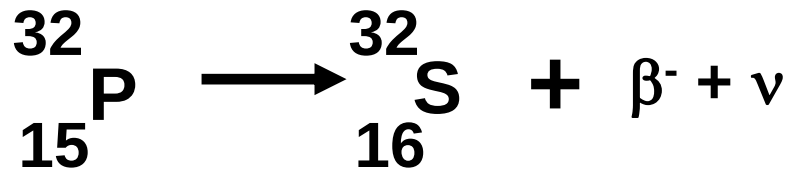
Beta Particle (β)

Daughter
Nucleus
Sulfur-32

Antineutrino



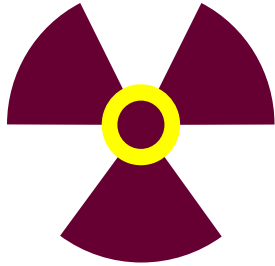
Parent Nucleus
Phosphorous-32



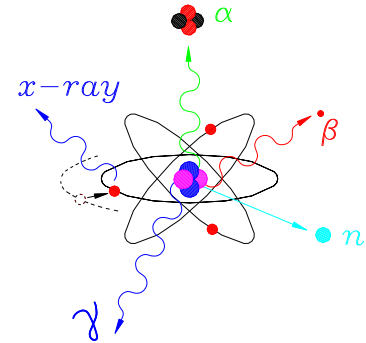
Beta Particle

Radiation Dose – VS- Radioactivity

Gamma Exposure Rate Approximations



$$X = \frac{6CEn}{d^2}$$



WHERE: **X** = **R/hr** (*gamma radiation exposure in Roentgen per hour*)

C = Activity in **Ci** (*amount of **radioactive material** present*)

E = Energy of gamma radiation in MeV

n = percent of occurrence per decay

d = **Distance** from the source in Feet

Isotopic Approximations:

- For ^{137}Cs : Exposure Rate = 0.33 R/hr/Ci @ 1 meter
- For ^{60}Co : Exposure Rate = 1.33 R/hr/Ci @ 1 meter

Units of Radiation Exposure & Dose

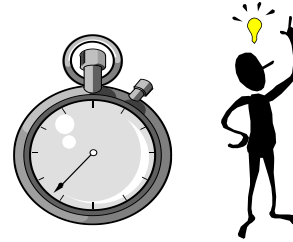
- **Roentgen (R):** unit of **exposure in air**
where $1 \text{ R} = 2.58 \times 10^{-4} \text{ coulombs/kg of air}$
- **RAD:** unit of **absorbed dose**
where $1 \text{ RAD} = 100 \text{ ergs/gram} = 6.2 \times 10^7 \text{ MeV/gram} = 9.93 \times 10^{-6} \text{ Watt-sec/gram}$
- **REM:** unit of **biological dose equivalent**
where $\text{REM} = \text{Quality Factor (QF)} \times \text{RAD}$

Radiation Type	Quality Factor (QF)
γ & X rays	1
β^-	1 – 1.7
Thermal Neutrons	3
Fast Neutrons	10
Heavy Ions	20

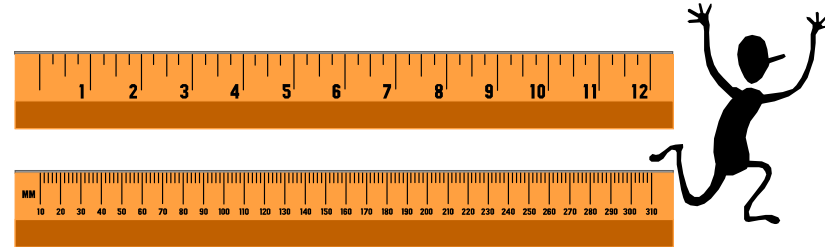
Approximation for gamma rays: 1 Roentgen ~ 1 RAD ~ 1 REM

Radiation Dose depends on:

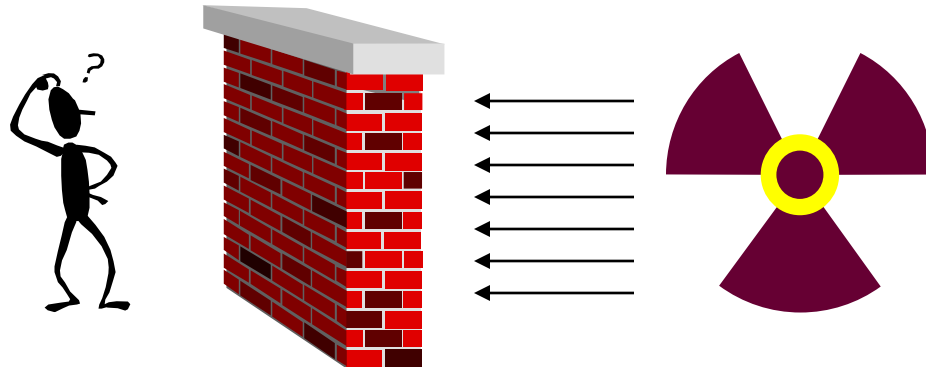
- TIME - time near source



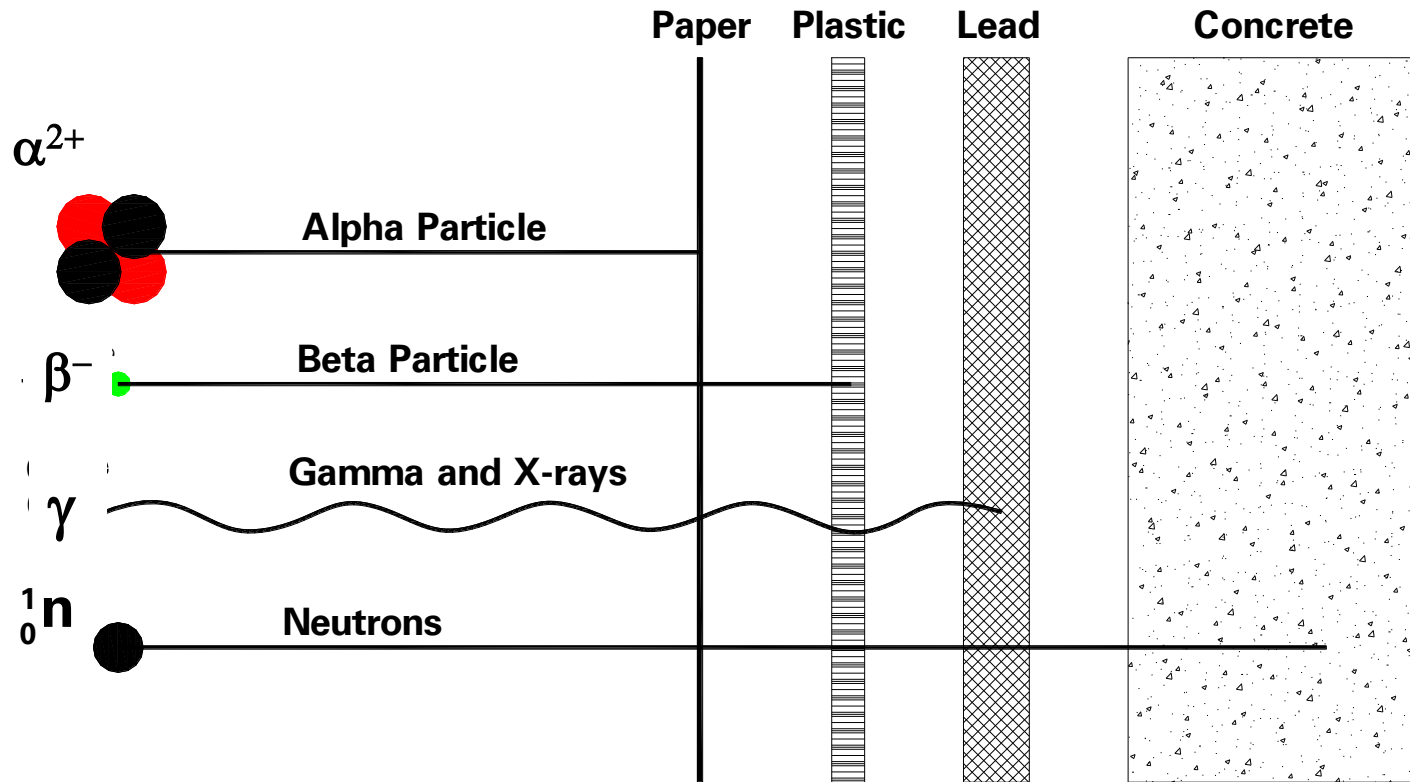
- DISTANCE - $DR_2 = DR_1 \left(\frac{d_1}{d_2} \right)^2$



- SHIELDING - Absorbs energy

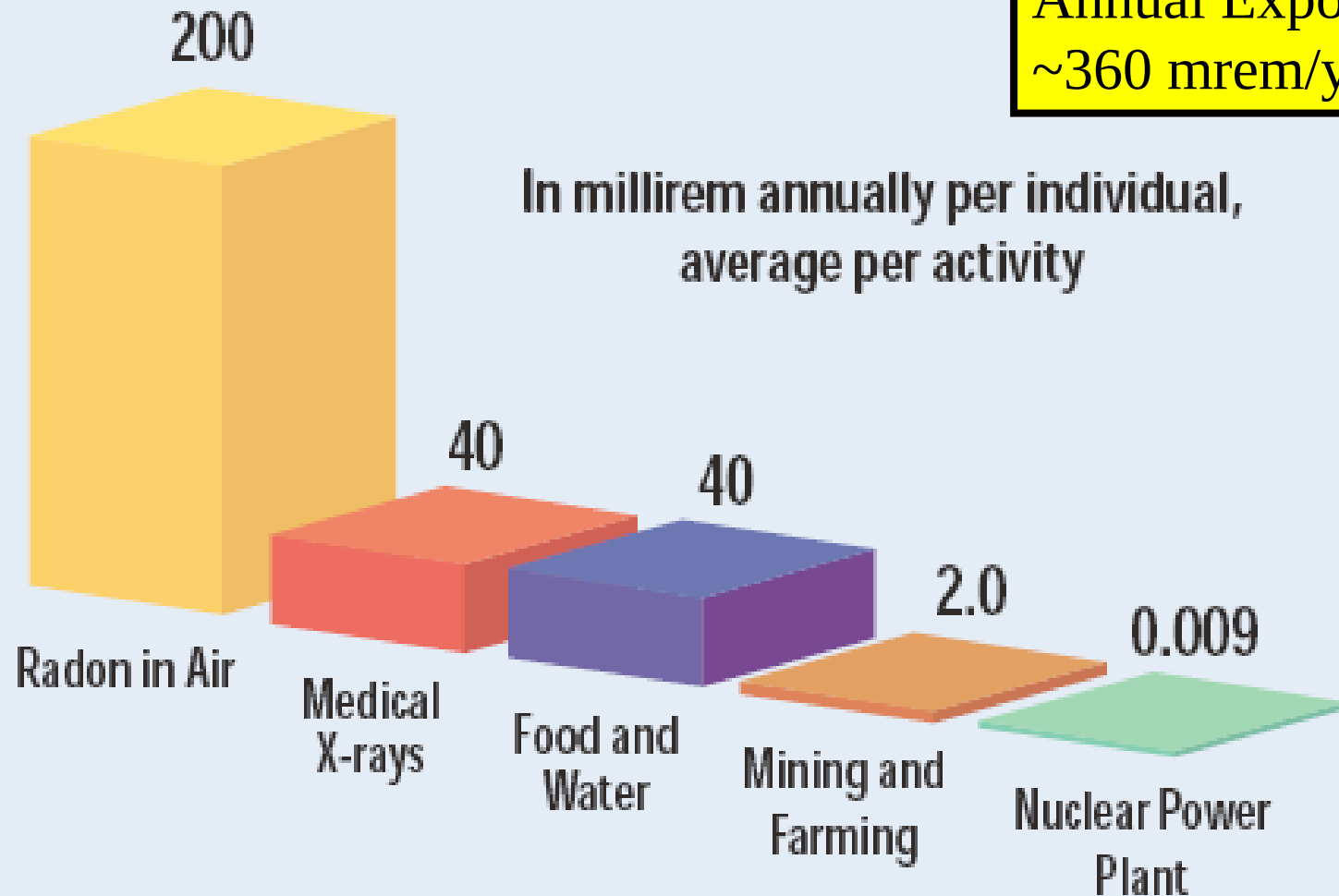


Penetrating Power of Radiation



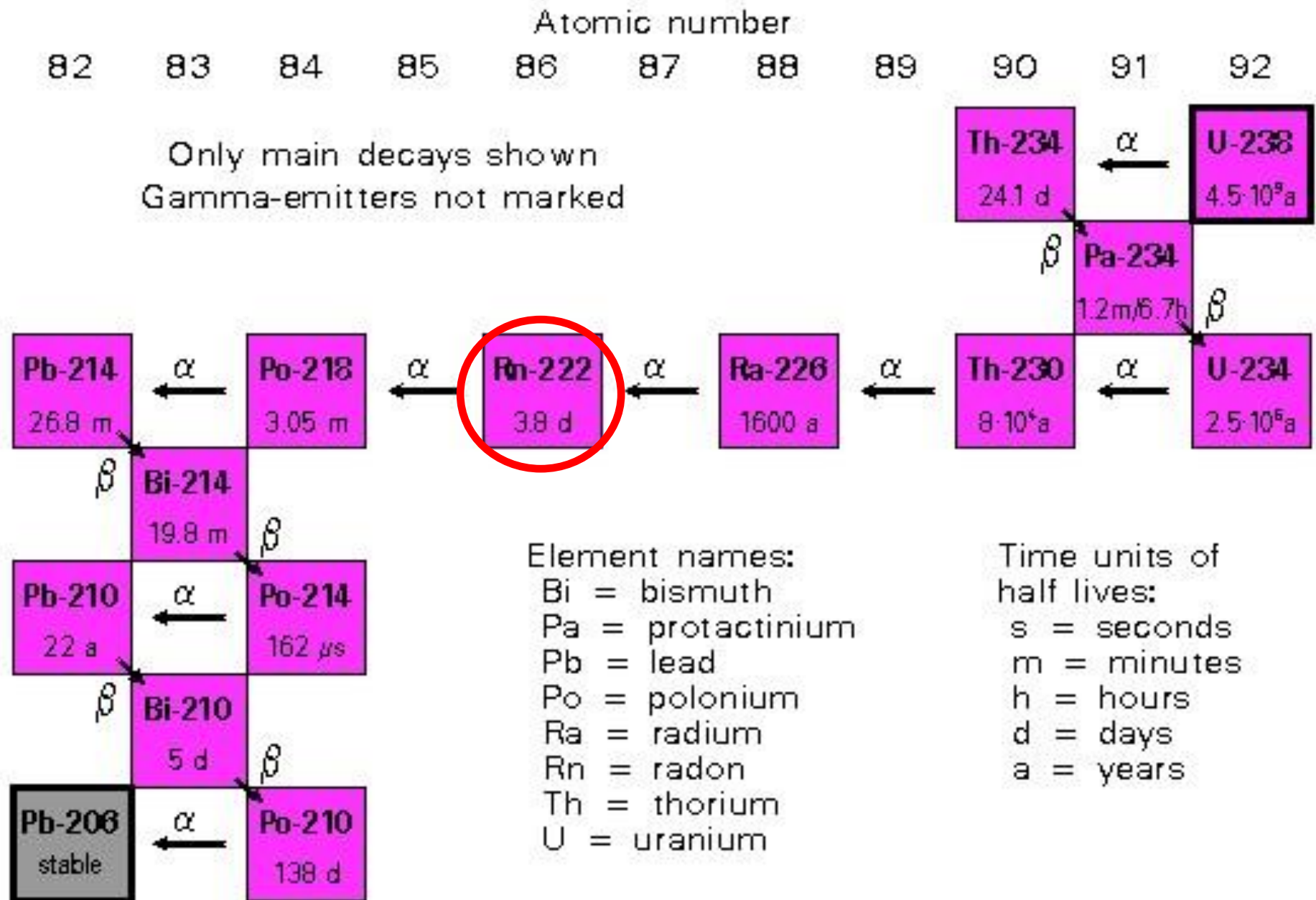
Radiation Exposure

Annual Exposure:
~360 mrem/year



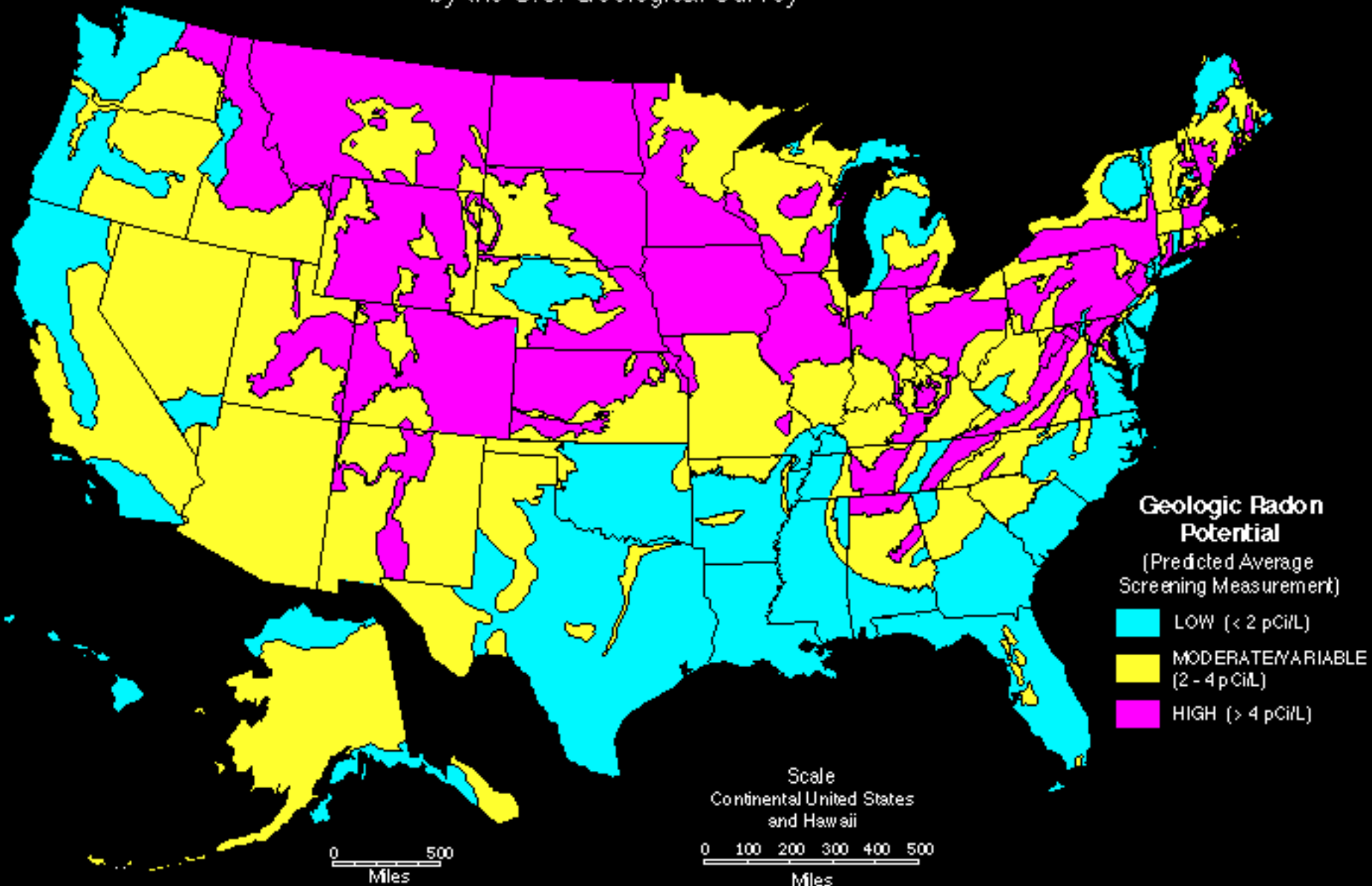
Sources: U.S. Environmental Protection Agency / National Council on
Radiation Protection and Measurements

The uranium-238 decay chain



GENERALIZED GEOLOGIC RADON POTENTIAL OF THE UNITED STATES

by the U.S. Geological Survey



Average Annual Radiation Dose for Various Occupations

<u>Occupation</u>	<u>Dose</u> (mrem/yr)
Airline flight crew member	1,000
Nuclear Power plant worker	700
Grand Central Station worker	120
Medical Personnel	70
PULSTAR Operator	<50



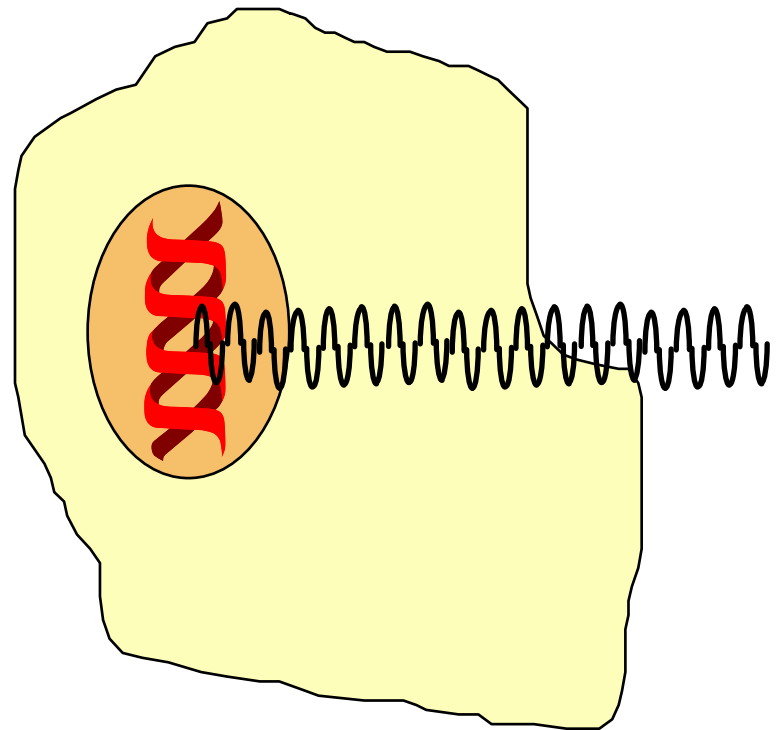
Biological Effects of Ionizing Radiation



Ionization disrupts molecular bonds in cells & tissues



- Rapidly reproducing cells at greatest risk:
 - Bone marrow
 - Fetus/Reproductive cells
 - Skin cells
 - Gastrointestinal tract
 - Lining of Lungs
- Highly specialized cells less at risk:
 - Muscles
 - Nerves



Biological Effects

Human data from WWII atomic bomb survivors and radiation accidents (e.g. weapons complex, Chernobyl)

- **Non-Stochastic (deterministic) effects** - **Severity** increases with dose
 - Skin reddening (burns)
 - Sterility
 - Acute Radiation Syndrome (ARS) (>150 REM):
 - Hematopoietic Syndrome – damage to blood & blood forming organs @ 100-1000 REM. White blood cells and blood forming stem cells in bone marrow are most sensitive and die. Infections, lack of oxygen transfer result.
 - Gastrointestinal Syndrome – damage to lining of stomach & intestines @ 1000-5000 REM. Lining of GI tract degrades/dies resulting in dehydration and blood poisoning.
 - Central nervous system – breakdown of nerve cell function. @ Doses > 5000 REM, death within a few hours due to fluid accumulation on brain.
- **Stochastic effects** - **Probability** of effect increases with dose:
 - Cancer
 - Genetic Mutations

Radiation Safety: Regs & Limits

Based on Linear No-Threshold Theory

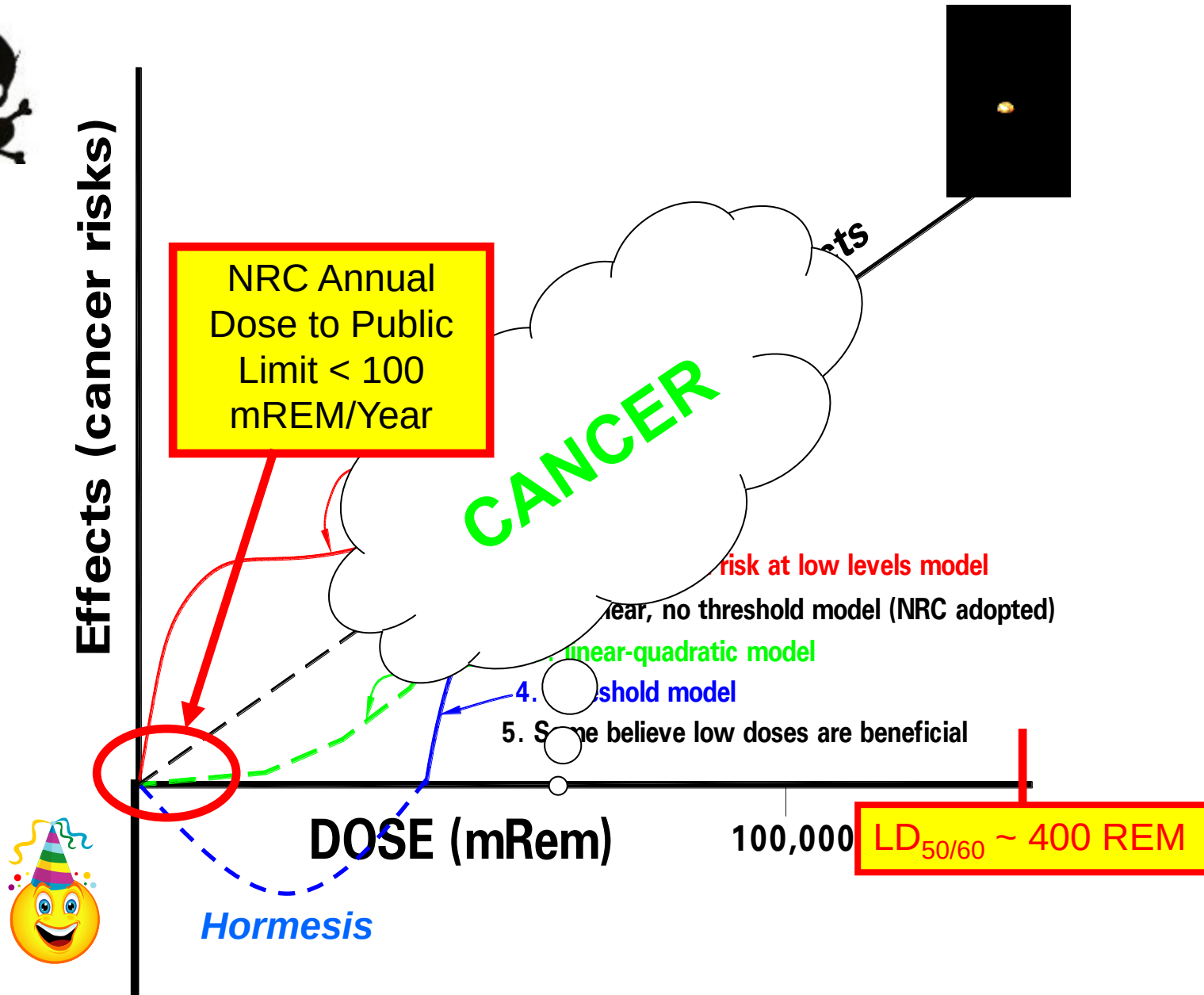
ALARA – Keep dose **As Low As Reasonably Achievable**

Where possible: **Reduce amount of radioactivity, minimize time near source, maximize distance from source, and utilize appropriate shielding.**

10CFR20 – NRC Radiation Protection Limits ([link](#))

- Radiation Workers :
 - ✓ **5 REM Whole Body**
 - ✓ **15 REM to Eye**
 - ✓ **50 REM to extremities**
- Public: **100 mREM/Year; 2 mREM in 1 hour**
- ALARA limits – 10% of NRC Limits

Radiation Dose (whole body) –vs- Risk



“Disasters” –vs- Disasters

Year	Location	Release / Exposure/ ‘Event’ ⁽¹⁾	# Immediate Deaths ⁽¹⁾	# Attributable Delayed Deaths ⁽¹⁾
1979	Three Mile Island (US)			
1984	Bhopal (India) – Union Carbide Chemical Plant			
1986	Chernobyl (Ukraine)			
2011	Fukushima (Japan)			
2011	Tōhoku (Japan)			

(1): Source - Wikipedia

“Disasters” –vs- Disasters

Year	Location	Release / Exposure/ ‘Event’ ⁽¹⁾	# Immediate Deaths ⁽¹⁾	# Attributable Delayed Deaths ⁽¹⁾
1979	Three Mile Island (US)	1.3x10 ⁶ Curies (Ci) 8-100 mrem max exposure		
1984	Bhopal (India) – Union Carbide Chemical Plant	Methy Isocyanate gas		
1986	Chernobyl (Ukraine)	>30% of core rad. mat. inventory >10 ⁸ Ci = yikes!		
2011	Fukushima (Japan)	Partial meltdowns in ~3 reactors and fuel pools		
2011	Tōhoku (Japan)	Earthquake & Tsunami		

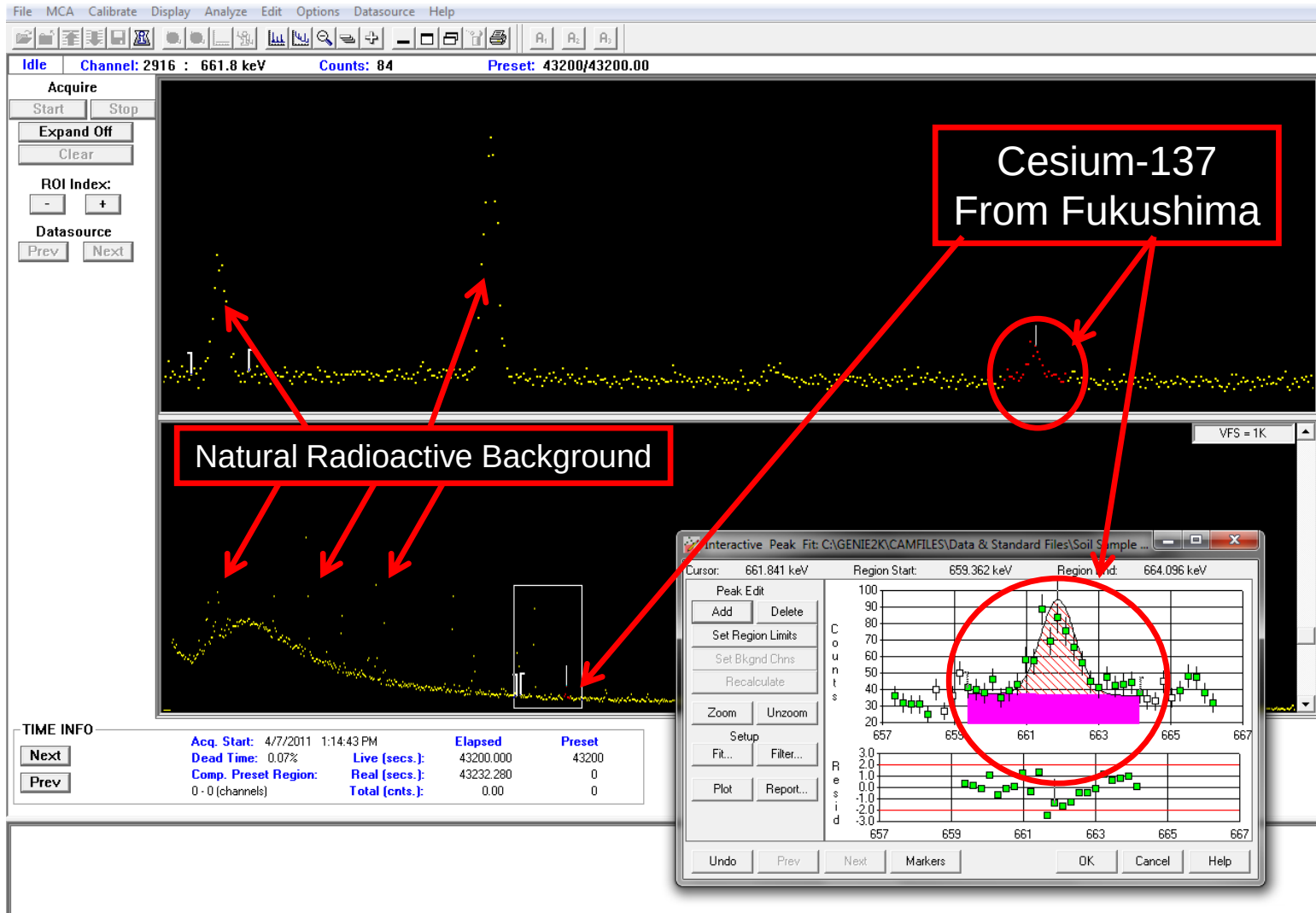
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1979	Three Mile Island (US)	1.3x10 ⁶ Curies (Ci) 8-100 mrem max exposure	0	0 – 2 cancers
1984	Bhopal (India) – Union Carbide Chemical Plant	Methy Isocyanate gas	3800 suffocated	8000 died later due to gas exposure
1986	Chernobyl (Ukraine)	>30% of core rad. mat. inventory >10 ⁸ Ci = yikes!	31	64 to 200,000 Cancers <i>(still hotly debated)</i>
2011	Fukushima (Japan)	<i>Partial meltdowns in ~3 reactors and fuel pools (oops)...</i>	0	?
2011	Tōhoku (Japan)	Earthquake & Tsunami	15,561 Dead 5,690 Injured	5313 Missing

(1): Source - Wikipedia

Sample taken Thursday April 7th, 2011 from storm drain in NCSU Brickyard



Summary

1. **Nuclear reactions** occur inside reactor which release radiation, either promptly or through decay of radioactive products.
2. **Nuclear radiation (α, β, γ, n)** emitted have different ranges and interactions with matter.
3. **Biological radiation dose** & resulting biological effects are dependent on type of radiation, duration and amount of exposure.
4. **10CFR20** provides regulatory limits for radiation exposure – *keep doses ALARA*.
5. Health risk is a function of biological dose.